

Introduction to robotic - SCCS 2023/24

Grade 3-5

Welcome to the Intro to robotic class, where you can learn how to build and program your own robots with VEX GO! VEX GO is a fun and easy way to explore STEM (science, technology, engineering and math) concepts through hands-on activities. You will use colorful parts, motors, sensors and a code base to create robots that can move, sense and interact with the world. You will also learn how to use VEXcode, a simple drag-and-drop coding software that lets you control your robots. In this club, you will work in teams to complete different challenges and projects, such as making a robot that can navigate a maze, play soccer or clean up trash. You will also have the opportunity to showcase your creations and compete with other clubs in friendly tournaments. By joining the Robotic class, you will develop your creativity, problem-solving and teamwork skills while having fun with robots. No prior experience is required, just bring your curiosity and enthusiasm. Are you ready to GO?

Students should bring their own Vex go kit to class.

VEX GO Education Kits cost \$199 on the official website.

<https://www.vexrobotics.com/go-kits.html>

Instructor: Xiaochen Li

Course syllabus:

1. Intro to building: Explore the VEX GO Kit to learn the names and functions of key pieces by building structures to take on a voyage to Mars.
2. Helping hand: Investigate and modify the Adaptation Claw to help solve authentic problems.
3. Animal Rescue: Explore building techniques to design and build a Ladder, Raft, and Mobile Rescue Vehicle to save the day.
4. Fraction: Explore equivalent fractions using the Fractions build and your VEX GO Kit pieces to compare fractions by size.
5. Fun with formation: Explore landform formation by building models of canyons, sand dunes, and deltas with your VEX GO Kit!
6. Super car: Investigate the variables that affect the movement of the Super Car.
7. Codebase: Use the Code Base and its sensor iterations to explore robot behaviors, controls, commands, and sensors through navigation challenges.
8. (Optional) Robot arm: Work with the Robot Arm, and its adaptations to learn what robotic arms can do and how they work!